

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

DEPARTMENT OF MATHEMATICS

PROBLEM SHEET 1

MA 2130 : Complex Variables

Year: 2026

1. Express the following complex numbers in the form of $x + iy$:

(a) $(-1 + i\sqrt{3})^{90}$,

(b) $(1 + i)^{11}$.

2. For any $z_1, z_2 \in \mathbb{C}$, prove the following identities

i) $|1 - \bar{z}_1 z_2|^2 - |z_1 - z_2|^2 = (1 + |z_1||z_2|)^2 - (|z_1| + |z_2|)^2$,

ii) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$. (**Parallelogram Identity**).

3. Describe all $z \in \mathbb{C}$ satisfying

$$|z - 1| + |z + 1| = 2.$$

4. What is the polar form of the following expression?

$$\frac{1 + \cos \phi + i \sin \phi}{1 + \cos \phi - i \sin \phi} \quad \left(0 < \phi < \frac{\pi}{2}\right).$$

5. Find all the fourth roots of $256i$.

6. Let $A := \{z \in \mathbb{C} : |z| < 1\}$ and $B := \{z \in \mathbb{C} : |z - 3| = 2\}$. What are all the accumulation points of $A \cup B$?

7. Give an example to show that an arbitrary intersection of open sets in $\mathbb{C}$ need not be open in $\mathbb{C}$. Similarly, show that an arbitrary union of closed sets in $\mathbb{C}$ need not be closed in $\mathbb{C}$.

8. For each of the following statements, state whether it is true or false. Justify your answer with a proof or a counterexample, as appropriate.

(i) The set $S = \{(x, 0) : x \in \mathbb{R}\} \cup \{(0, y) : y \in \mathbb{R}\}$ is polygonally connected,

(ii) A closed set can never be a domain.

9. Describe the geometric interpretation of operations on complex numbers using the following examples. Support your explanation with appropriate figures for clarity. In each case (except the last three), clearly specify the type of geometric transformation involved, such as translation, rotation (about the origin or another point), or scaling.

$$\text{Let } S := \{x + iy \in \mathbb{C} \mid -1 \leq x \leq 1, -1 \leq y \leq 1\},$$

$$zS := \{zs \mid s \in S\},$$

$$z + S := \{z + s \mid s \in S\}$$

for all $z \in \mathbb{C}$. Then describe

- (a) $5S$
 - (b) $\frac{1}{2}S$
 - (c) $e^{\frac{i\pi}{4}}S$
 - (d) $3e^{\frac{i\pi}{4}}S$
 - (e) $(1+2i) + e^{\frac{i\pi}{4}}S$
 - (f) $(1+2i) + e^{\frac{i\pi}{4}}S$
 - (g) $(1+2i) + e^{\frac{i\pi}{4}}S$
 - (h) $2e^{\frac{i\pi}{4}}(4+S)$
 - (i) rotate S about $z = 6$ by an angle of θ
 - (j) the range of $f(t) = te^{it}$, $t \in [0, 8\pi]$
 - (k) the range of $g(t) = (5 + \sin t)e^{it}$, $t \in [0, 2\pi]$
10. Show that a nonempty subset $S \subseteq \mathbb{C}$ which is both open and closed must be equal to $\mathbb{C}$. (Hint: For $x \in S$ and $y \in \mathbb{C}$, show that every point in the line segment joining x, y are also in S .)
 11. Prove that the set of all interior points of a subset $S \subseteq \mathbb{C}$ is an open set.
 12. Can there exist a non-empty proper subset S of $\mathbb{C}$ such that the boundary set of S is empty? Justify your answer with a proof or a counterexample, as appropriate.